FreshSaver

SMART FOOD TRACKING, RECIPE GENERATION & FOOD RESCUE PLATFORM

Comprehensive Functionality & Architecture Guide | Version 2.0 | Production Release

FreshSaver is a full-featured, zero-waste grocery intelligence system engineered to eliminate household food waste. The platform provides real-time barcode and receipt scanning, intelligent shelf-life tracking, automated expiry notifications, generative AI meal creation prioritizing at-risk ingredients, direct community food bank donations, smart shopping lists with automatic inventory transformation, and a competitive community leaderboard offering real savings discounts.

Executive Overview of Application Pillars

1. Real-Time Pantry Inventory & Expiration Engine

CORE ENGINE

- **Smart Lifecycle Badging:** Calculates dynamic expiration windows with color-coded urgency states (Expired, Urgent <= 3 days, Safe).
- **Granular Organization:** Filter and sort inventory by category (Produce, Dairy, Meat, Pantry, Bakery, Freezer) and physical storage location.
- **Instant Actions:** One-tap controls to mark food consumed, donate directly to local charities, adjust quantities, or edit shelf life.

2. Multi-Mode Scanner: Barcodes, Receipts & Quick Add

SCANNER

- **Camera Barcode Scanner:** Browser BarcodeDetector API with real-time continuous video preview and Open Food Facts API product lookup.
- **OCR Receipt Parser:** Ingests itemized supermarket receipts and extracts line items, quantities, and auto-assigns category shelf lives.
- **Custom Manual Entry:** Includes pre-configured food category expiration presets and manual date overrides for farmers-market goods.

3. Zero-Waste AI Recipe Studio (Powered by Gemini AI)

AI STUDIO

- **Expiring Ingredients First:** Prioritizes pantry items nearest to expiration to prevent spoilage before it happens.
- **Dietary Customization:** Customized recipes for Vegan, Vegetarian, Gluten-Free, High-Protein, Quick (<20m), and Kid-Friendly diets.
- **Pantry Match Score:** Analyzes which ingredients you have and lists missing staples with one-click additions to your shopping list.

1. Pantry Lifecycle & Expiration Monitoring

Automated freshness tracking, days-left countdowns, and storage management.

The Pantry module acts as the central command center for all food items currently owned by the user. Each food entry maintains rich metadata including date added, estimated expiry date, calculated days remaining, quantity, unit type, category, and storage zone.

- **Dynamic Expiry Formula:** Days remaining are calculated daily relative to midnight local time. The system categorizes items into three critical tiers:
- • Red (Expired or Expiring Today): Items requiring immediate use or compost logging.
- • Amber (Urgent - 1 to 3 Days): High-priority items automatically flagged for AI recipe generation and donation suggestions.
- • Emerald (Fresh - 4+ Days): Standard items safely stored with plenty of shelf life remaining.
- **Physical Storage Zones:** Supports distinct separation between Fridge, Freezer, and Pantry/Dry Storage to accommodate differing shelf-life realities.
- **Quick Filtration & Search:** Instant search query matching item names, plus category chips (All, Produce, Dairy, Meat, Bakery, Pantry, Frozen).
- **Lifecycle State Transitions:** Items can transition into Consumed (earning food saver XP), Donated (generating donation manifest), or Removed.

2. Multi-Mode Scanner System

Hardware camera barcode scanning, receipt OCR, and manual entry.

FreshSaver eliminates tedious manual food entry by providing three unified input methods designed for speed and convenience.

Camera Barcode Scanning

- **Hardware Camera Access:** Requests standard video capture with autofocus and environment (rear) camera preference.
- **Native BarcodeDetector Integration:** Leverages modern browser hardware acceleration supporting EAN-13, EAN-8, UPC-A, UPC-E, and QR codes.
- **Product Data Resolution:** Scanned barcodes query the global Open Food Facts API database, retrieving official brand names, food categories, and package weights.
- **Automatic Expiration Projection:** Assigns realistic baseline expiration dates based on recognized categories (e.g. Dairy defaults to 7 days, Fresh Meat to 4 days, Pantry dry goods to 90 days).

Smart Supermarket Receipt Parser

- **Line-by-Line Tokenizer:** Users can paste or capture raw text from digital receipts, supermarket loyalty emails, or delivery summaries.
- **Receipt Cleanup Heuristics:** Strips out store codes, dates, transaction IDs, tax lines, and currency amounts ($0.00) to isolate genuine food names.
- **Batch Ingestion:** Adds 10 to 30 items to the pantry inventory in a single click, saving users 15+ minutes of manual entry per shopping trip.

3. Generative AI Recipe Studio

Zero-waste culinary intelligence powered by Google Gemini AI.

Food waste often occurs because households do not know how to combine disparate, half-used ingredients. The AI Recipe Studio analyzes your unique pantry inventory and constructs tailored, appetizing recipes designed specifically to rescue expiring goods.

Gemini AI Recipe Generation Engine

AI INTEGRATION

- **Pantry-Aware Prompts:** Injects urgent items expiring within 3 days into the system prompt with explicit instructions to prioritize them.
- **Dietary & Lifestyle Filters:** Enforces user preferences: Vegetarian, Vegan, Gluten-Free, High-Protein, Quick & Easy (<20 mins), and Family-Friendly.
- **Structured Recipe Output:** Generates detailed step-by-step instructions, prep and cook timings, difficulty rankings, estimated calories, and pantry utilization scores.
- **Missing Ingredients Sync:** Identifies missing pantry staples and allows users to push missing items directly into their Shopping List with one tap.
- **Offline Recipe Library:** Includes a robust curated offline fallback database with smart ingredient matching if internet access or API connectivity is temporarily unavailable.

4. Local Food Donation & Community Rescue

Connecting surplus edible food to local food banks and community fridges.

When households realize they have surplus edible food that will not be consumed in time, FreshSaver facilitates seamless donation to verified local charities, emergency food pantries, and community fridges.

- **Verified Community Directory:** Built-in directory of community food organizations featuring street addresses, hours of operation, phone numbers, and distance markers.
- **Accepted Foods Filter:** Clear guidelines indicating which organizations accept fresh produce, sealed dairy, unexpired canned items, or prepared meals.
- **Digital Donation Manifest:** Select expiring items directly from your pantry to generate an itemized digital receipt/manifest to hand over to volunteers upon drop-off.
- **Impact Tracking & Rewards:**Every item donated awards bonus Food Saver XP and community tier points, reflecting tangible community impact.

5. Smart Shopping List with Auto-Pantry Conversion

Seamless cycle from planning to purchasing to pantry tracking.

FreshSaver closes the food lifecycle loop through an interactive shopping list. Instead of re-typing groceries when you get home from the store, the list directly promotes bought items into active pantry tracking.

- **One-Tap Auto-Pantry Transition:**Checking off an item prompts the user or automatically converts the grocery item into a live pantry item.
- **Pre-Assigned Expiry Presets:** Items inherit category shelf lives upon purchase, eliminating manual date calculation.
- **Recipe Integration:** Add missing ingredients directly from any generated recipe with a single tap.
- **Item Grouping & Sorting:** Organized by supermarket aisles (Produce, Dairy, Meat, Frozen, Pantry) for efficient in-store navigation.

6. Top Savers Leaderboard & Discount Rewards

Gamified motivation and tangible economic incentives for reducing food waste.

To build lasting habits, FreshSaver incorporates a motivating tier and reward system that celebrates food waste reduction and unlocks real-world partner discounts.

Community Tier Hierarchy & Real Discounts

REWARDS TIER

- **Bronze Saver (0 - 249 pts):** 5% off select local organic grocery partners & bulk food bins.
- **Silver Saver (250 - 599 pts):** 10% off eco-friendly grocery partners, farm stands & bakery surprise bags.
- **Gold Saver (600 - 1,199 pts):** 15% off fresh market partners + free zero-waste food storage container kit.
- **Zero-Waste Champion (1,200+ pts):** 20% off all participating green partners, featured spotlight, and VIP community badge.

- **Environmental Impact Metrics:** Tracks total estimated CO2 emissions avoided (kg), edible meals rescued, and household dollars saved ($).
- **Community Leaderboard:** Monthly ranking displaying top local food savers with badges, streak counters, and peer encouragement.

7. Security, User Auth & Cross-Device Sync

Privacy-first architecture with multi-device synchronization.

- **Flexible Auth Flow:** Supports Email/Password registration, instant login, and seamless guest mode for immediate evaluation.
- **Local-First Resilience:** All user data (pantry items, shopping lists, points, settings) caches reliably in browser local storage and syncs upon connection.
- **Data Sovereignty & Export:** Full JSON export and import capabilities, allowing users to backup their complete pantry and donation history at any time.
- **Clean UI Architecture:** Adheres to strict anti-slop guidelines: no debug overlays, high-contrast typography, and intuitive mobile ergonomics.

8. Progressive Web App & Android App Download

Standards-compliant PWA, WebAPK, and GitHub Actions Android compilation.

FreshSaver is engineered as both a responsive modern web application and an installable native-like mobile application for Android and iOS.

Download & Distribution Specifications

DISTRIBUTION

- **Standards-Compliant Web Manifest:** Includes high-res 192x192, 512x512, and maskable icons, standalone display mode, and theme color #059669.
- **Offline Service Worker (PWA):** Powered by Workbox & vite-plugin-pwa, caching core assets, icons, fonts, and application bundles for offline usage.
- **Trusted Web Activity (TWA):** Pre-configured twa-manifest.json enabling automated packaging into native Android APK / AAB bundles via Google Bubblewrap.
- **Automated GitHub Action:** .github/workflows/build-apk.yml compiles the Android APK directly on every release tag, making release APKs downloadable via GitHub Releases.

SUMMARY OF CORE APPLICATION CAPABILITIES

Feature Module	Primary User Capability	Technology Used
Pantry Tracker	Tracks shelf life, expiry countdowns, storage zones	React + TypeScript + LocalStorage
Barcode Scanner	Live camera scan, barcode lookup via Open Food Facts	BarcodeDetector API + HTTP Fetch
Receipt OCR	Extracts and parses itemized supermarket receipts	Regex Line Tokenizer + Category Classifier
AI Recipe Studio	Cooks meals prioritizing at-risk ingredients	@google/genai (Gemini AI API)
Food Donation	Finds local food banks, builds donation manifests	Geolocated Org Registry + Manifest Export
Smart Shopping	Aisle-sorted list that auto-converts to pantry items	State Machine + Shelf-life presets
Top Savers	Points, tiers, CO2 stats & real grocery discounts	Gamification Logic + Partner Promo Engine
Android & PWA	Installable on home screen, offline cache, APK build	vite-plugin-pwa + Bubblewrap + GH Actions